Rational Exponents

Page 2 — Building Intuition

Recall:

$$2 + 2 + 2 + 2 + 2 = 5 \times 2$$

$\Rightarrow$ Multiplication is repeated addition

$$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 3^6$$

$\Rightarrow$ Exponent is repeated multiplication

My Favourite Exponent:

$$\underbrace{2^{10}}_{\text{base}} = \underbrace{2 \cdot 2 \cdots 2}_{10} = 1024 \quad \leftarrow \text{product}$$

Type	**Goal**	**Example**
Type 1 (learned)	Calculate the *product*	$2^{10} =$ ____
Type 2 (today)	Find the *base*	$\square^{10} = 1024$
Type 3 (future)	Find the *power/exponent*	$10 = \log_2 1024$

Page 3 — Finding the Base

Analogy with multiplication:

$$2 \times 3 = 6 \quad \Rightarrow \quad \square \cdot 3 = 6 \quad \Rightarrow \quad \square = 6 \div 3 \quad \text{(long division)}$$

Same idea for exponents:

$$2^{10} = 1024 \quad \Rightarrow \quad \square^{10} = 1024$$

$$\underbrace{x \cdot x \cdot x \cdot x \cdots x}_{10} = 1024$$

Answer:

$$\boxed{2 = \sqrt[10]{1024}}$$

Page 4 — Radical Notation

We introduce the **radical sign**:

$$\sqrt[n]{a}$$

n : **index**

a : **radicand**

Page 5 — Inverse Relationship

Recall from multiplication:

$$3 \cdot 4 = 12 \quad \Rightarrow \quad 4 = 12 \div 3 \quad \Rightarrow \quad 3 = 12 \div 4$$

Same idea for exponents:

$$2^2 = 4 \quad \Rightarrow \quad 2 = \sqrt{4}$$

$$3^2 = 9 \quad \Rightarrow \quad 3 = \sqrt{9}$$

$$2^3 = 8 \quad \Rightarrow \quad 2 = \sqrt[3]{8}$$

$$3^3 = 27 \quad \Rightarrow \quad 3 = \sqrt[3]{27}$$

Convention:

$\sqrt{} = \sqrt[2]{}$ (index 2 is invisible — most commonly used root)

$2 = 2^1$ (similarly, exponent 1 is invisible)

Page 6 — Fill In

$$2^5 = \boxed{32} \quad \Rightarrow \quad 2 = \sqrt[5]{32}$$

$$3^4 = \boxed{81} \quad \Rightarrow \quad 3 = \sqrt[4]{81}$$

$$4^2 = \boxed{} \quad \Rightarrow \quad 4 = \sqrt{}$$

$$2^{10} = \boxed{} \quad \Rightarrow \quad 2 = \sqrt[10]{}$$